

ODTUG
KSCOPE 26

The background is a stylized illustration of a mountain landscape. In the foreground, a hiker in silhouette with a large backpack and a cap walks along a winding path. To the left, a brown, Bigfoot-like creature stands on a grassy slope. In the middle ground, a city skyline with several colorful buildings (orange, green, and blue) is visible. The background features blue mountains and a large yellow sun in an orange sky.

DENVER · JUNE 14 -18

Building Secure EPM Pipelines with OAuth

From REST APIs to Multi-Cloud Integration

Demo-Driven

Technical Deep Dive

Live Examples

Presented by Dayalan - RSM

Modernize • Automate • Secure



What This Session Will Deliver



Real-World OAuth Application

Demonstrates practical token-based authentication using OAuth in Oracle EPM REST API integration scenarios.

Live Demonstrations

Focuses on live demos showing secure connection setup and maintenance in real use cases.

CI/CD Pipeline Automation

Illustrates OAuth-enhanced CI/CD pipelines automating data refresh, scenario execution, and monitoring securely.

Multi-Cloud Integration & Migration

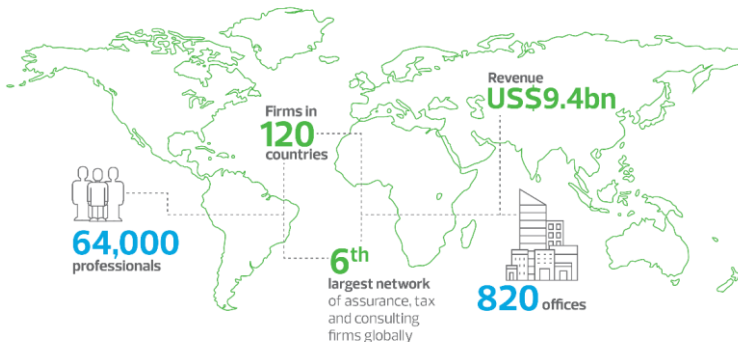
Explores secure multi-cloud patterns and migration from Basic Authentication to OAuth for modernized systems.

RSM Firm Overview

As the **5th largest firm** in the U.S., with over **17,000 professionals** in **79 cities** in the U.S. and six locations in Canada, four in India and one in El Salvador, we deliver assurance, tax and consulting services to our clients. As we prepare to celebrate our **100th anniversary in 2026**, we have evolved our strategy to solidify our leadership as the first-choice advisor to middle market leaders globally.

Our approach to client service is rooted in deep industry knowledge and technical acumen. Our first-choice advisors do not just provide services, they bring diverse perspectives, innovative ideas and data-driven insights to our clients.

RSM International is a **global network** of independent assurance, tax and consulting firms.



RSM industry solutions and capabilities



How we differentiate



RSM's service model. Our service team model emphasizes continual involvement by partners and senior managers. This means you will work with senior members in a hands-on relationship to receive thought leadership, idea sharing, open communication, proactive planning, and candid and lively dialogue.



RSM's drive for innovation. A focal point of our audit, tax and consulting practice is the drive for constant innovation. Through innovation, we are committed to bringing efficiency to processes that limit disruption and unlock deeper insights that help our clients move forward confidently.



RSM's industry analysts. The RSM Industry Eminence Program is designed to help you navigate the speed and scope of the ongoing digital revolution while providing forward-looking industry insights to our clients. Our clients have access to senior analysts equipped to understand, forecast, and communicate economic, business and technology trends shaping the nonprofit industry.



RSM's value proposition. Our industry-focused professionals possess the depth and breadth of expertise expected from a leading professional services firm, delivered through a client service model that employs a more hands-on, collaborative approach than other larger firms.

RSM's Breadth of Service



Consulting

- Application and infrastructure support
- Business intelligence
- Business process improvement
- Cost optimization
- Finance and accounting outsourcing
- Information security
- Internal control consulting
- IT strategy and optimization
- Lease due diligence
- Operations and supply chain
- People and organization optimization
- Project management
- SOX compliance
- Systems integrations
- Systems selections
- System Implementations



Audit

- Asset valuation
- Complex debt and equity transactions
- Derivatives (hedging)
- Discounting and other promotional activities
- Exclusivity agreements/purchase commitments
- Gift card accounting and related breakage
- Incentive compensation arrangements
- Lease accounting
- Loyalty programs
- Self-insurance



Tax

- Credits and Incentives
 - EOTCs
 - Research and development
 - TIPS credits
 - WOTCs
- Expenditure analysis
 - Meals and entertainment
 - Charitable
 - Package design costs
 - Research and development
- Leasing
 - Anchor tenant benefits
 - Qualified leasehold improvements
- Tenant allowances
 - Bonus depreciation
 - Cost segregation
 - Retail depreciation
 - TARS
- Fixed assets
 - Sales and use
 - Structuring/planning
 - Unclaimed property/property tax
- State and local issues
 - Tax treaty/withholding comp.
 - Transfer pricing
 - VAT/GST
- International

WHO IS THIS SESSION FOR?

This session is designed for hands-on technical professionals



EPM Administrators

- Manage IDCS clients and scopes
- Replace Basic Auth across environments
- Monitor token usage and health



Solution Architects

- Design cross-cloud EPM architectures
- Define OAuth integration patterns
- Govern multi-system data flows



FP&A Technologists

- Automate planning cycles via REST
- Trigger financial consolidations
- Secure data pipeline orchestration



Integration Specialists

- Build AWS/Azure connectors
- CI/CD pipeline OAuth integration
- Third-party app authentication

HOW BASIC AUTHENTICATION WORKS IN ORACLE EPM

Understanding the mechanics behind username + password authentication

Basic Auth Mechanism

Every REST API call to Oracle EPM Cloud sends the user's credentials in the HTTP Authorization header — encoded (not encrypted) in Base64.

Authorization: Basic <Base64(username:password)>

Example:

Username: `epm_admin`

Password: `S3cur3P@ss!`

Encoded: `ZXBtX2FkbWluO1MzY3VyM1BAc3Mh`

Header: `Authorization: Basic ZXBtX2FkbW...`

⚠ Important: Base64 is reversible encoding — NOT encryption.

Anyone who intercepts the header can decode credentials instantly.

Basic Auth Request Flow

- 1 Client prepares username:password string
- 2 Base64 encodes the combined string
- 3 Sends encoded string in Authorization header with EVERY request
- 4 EPM Cloud server decodes and validates credentials
- 5 If valid, returns API response
- 6 Repeat for EVERY subsequent API call

Key Insight: Credentials are transmitted on EVERY single API call — dramatically increasing the attack surface compared to token-based auth.

```
C:\Oracle\EPM Automate\bin>epmautomate login dayalan.punniyamoorthy@rsmus.com hGn86n25Fb@ https://epm13-test-a641377.epm.us-phoenix-1.ocs.oraclecloud.com
```

Login successful

BASIC AUTH — LIMITATIONS IN DETAIL

Why Oracle recommends migrating away from Basic Authentication

Security Vulnerabilities

- X Credentials are Base64 encoded — easily reversible (not encrypted)
- X Username + password sent with EVERY API call — increasing exposure
- X Hard-coded passwords in scripts, config files, and batch jobs
- X No token expiry — compromised credentials remain valid indefinitely
- X Susceptible to replay attacks if HTTPS is not enforced
- X Shared service account passwords across teams = single point of failure

Functional Limitations

- X NOT supported with Multi-Factor Authentication (MFA)
- X Does NOT support SSO or federated identity login
- X Password changes instantly break ALL active integrations
- X Every script, pipeline & scheduled job must be updated manually
- X No fine-grained scope control — all-or-nothing access
- X No centralized token revocation — only option is password reset

Enterprise & Compliance Gaps

- X Limited audit trail — no token lifecycle tracking
- X Cannot support multi-cloud workflows natively
- X Breaks CI/CD automation patterns (secrets in pipelines)
- X No delegated authorization — every call uses full user identity
- X Difficult to meet SOC 2 / ISO 27001 compliance requirements
- X No programmatic way to rotate credentials automatically

Oracle Readiness Update (November 2025):

"Oracle recommends the use of OAuth 2.0 instead of Basic Auth to ensure enhanced security while using EPM Automate and EPM REST APIs."

Applies to: Planning, FCCS, FreeForm, Tax Reporting, Account Reconciliation, EDM, Narrative Reporting, EPCM

WHAT IS OAUTH 2.0?

The industry standard for secure, delegated authorization

OAuth 2.0 Defined

OAuth 2.0 is an **open authorization framework** that enables secure, token-based access to protected resources — without sharing credentials.

Instead of passwords, OAuth uses **short-lived access tokens** with defined scopes, issued by a trusted authorization server (IDCS).

Key Components:

Client App

Oracle EPM, script, or pipeline

Auth Server

Oracle IDCS issues tokens

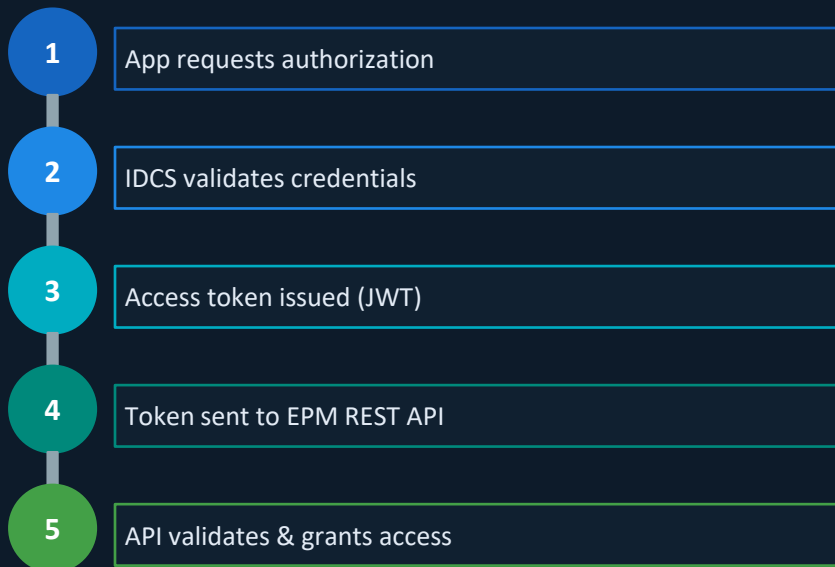
Resource Server

EPM REST API endpoints

Access Token

JWT bearer, time-limited

OAuth Token Flow



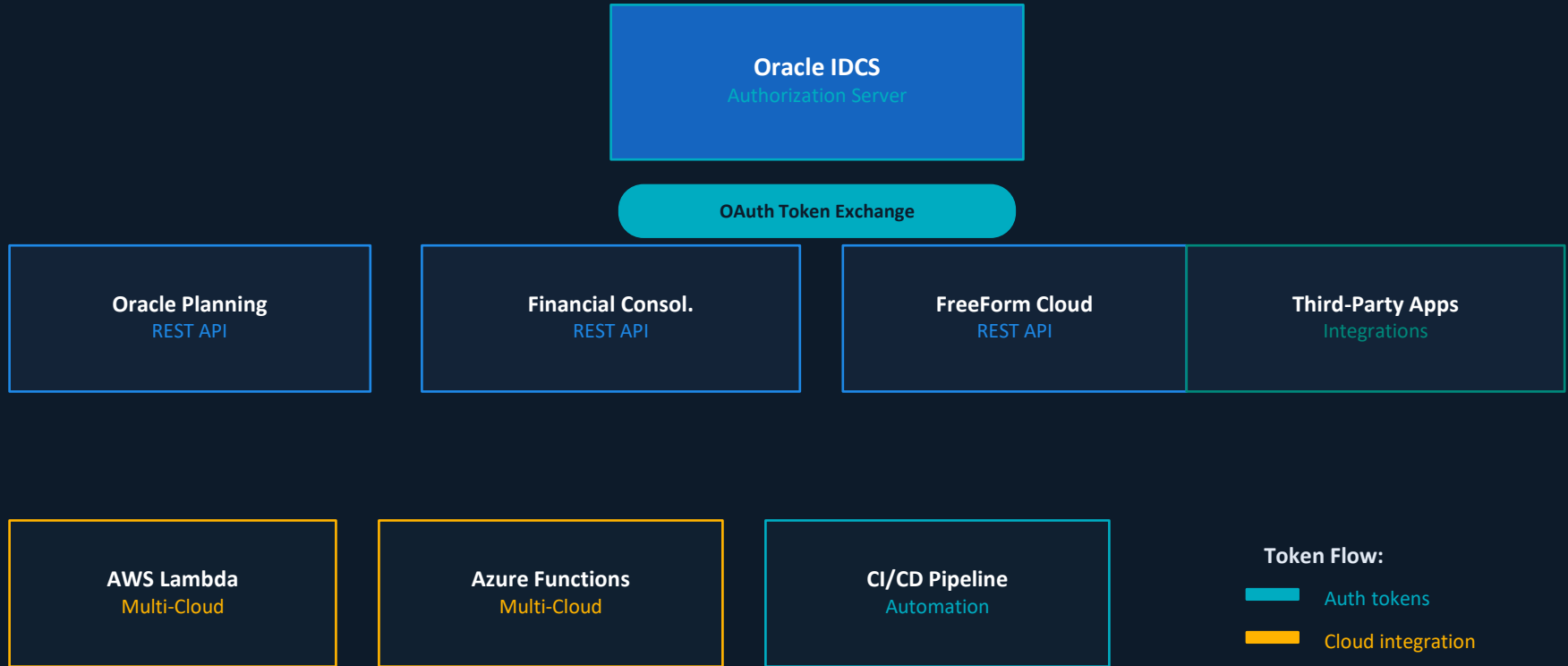
OAuth vs Basic Auth: A REAL-WORLD COMPARISON

Criteria	Basic Authentication	OAuth 2.0
Credential Type	✗ Username + Password	✓ JWT Access Token
Transmission	✗ Base64 encoded (insecure)	✓ Bearer token (signed)
Expiry	✗ Never expires	✓ Configurable TTL
Scope Control	✗ All-or-nothing access	✓ Fine-grained scopes
Revocation	✗ Password change only	✓ Instant token revocation
Multi-Cloud	✗ Not supported natively	✓ Full OAuth federation
Audit Trail	✗ Limited logging	✓ Full token lifecycle audit
CI/CD Pipelines	✗ Hard-coded secrets	✓ Secure client credentials

✓ RECOMMENDED FOR ENTERPRISE EPM

EPM OAUTH ARCHITECTURE OVERVIEW

End-to-end secure integration landscape



HOW ACCESS TOKENS WORK IN ORACLE EPM

The short-lived JWT bearer token that replaces passwords

Access Token Explained

An Access Token is a short-lived JWT (JSON Web Token) issued by Oracle Identity Cloud Service (IDCS). It acts as a "digital pass" that proves a client is authorized — WITHOUT ever sending a username or password.

Key Characteristics:

- **Format:** JWT (JSON Web Token) — digitally signed
- **Type:** Bearer token
- **Default Expiry:** 3600 seconds (1 hour)
- **Customizable:** Yes — via scope parameter
- **Contains:** User context, scopes, client info, expiry
- **Usage:** Sent in Authorization header with every API call

Authorization: Bearer eyJ4NXQjUzI...evRJChXTRfzn6WlCw

Getting an Access Token

- 1 Client sends refresh_token + client_id to IDCS
POST https://<IDCS-URL>/oauth2/v1/token
grant_type=refresh_token
- 2 IDCS validates the refresh token and client identity
- 3 IDCS issues a NEW access token + NEW refresh token
- 4 Client uses access token in API calls:
Authorization: Bearer <access_token>
- 5 When token expires → repeat from Step 1 (automated)

Response JSON:

```
{ "access_token": "eyJ...", "expires_in": 3600,  
  "refresh_token": "AAyy...", "token_type": "Bearer" }
```



Why JWT? The access token is self-contained — the EPM server can validate it locally without calling IDCS again. This makes API calls faster and reduces dependency on the authorization server.

HOW REFRESH TOKENS WORK IN ORACLE EPM

The long-lived token that keeps your automation running without passwords

Refresh Token Explained

A Refresh Token is a long-lived credential used to obtain NEW access tokens without re-authentication. Think of it as a "renewal pass."

Key Characteristics:

- **Default Expiry:** 604,800 seconds (7 days)
- **Maximum Expiry:** 7 days — CANNOT be extended
- **Rotation:** Each use generates a NEW refresh token
- **One-Time Setup:** First token requires browser interaction
- **Subsequent Use:** Fully automated — no user interaction
- **Storage:** Must be encrypted (.epw files)

First-Time Setup: Device Code Grant

- 1 POST to IDCS /oauth2/v1/device endpoint
response_type=device_code
scope=urn:opc:dm:myscopes__offline_access
- 2 IDCS returns device_code, user_code, verification_uri
{ "user_code": "SDFGHJKL", "expires_in": 300 }
- 3 User opens verification_uri in browser
→ Authenticates with IDCS/SSO credentials
→ Enters the user_code when prompted
- 4 POST to IDCS /oauth2/v1/token endpoint
grant_type=urn:ietf:params:oauth:grant-type:device_code
- 5 IDCS returns FIRST access_token + refresh_token
→ Store refresh_token securely (ONE-TIME manual step)

Refresh Token Rotation Strategy for Unattended Automation

Since refresh tokens expire after 7 days, create a scheduled job that refreshes the token every 6 days:

EPM Automate: epmautomate login ignoredUsername OAuth2.epw https://epm-host → epmautomate logout

REST APIs: POST grant_type=refresh_token → save new refresh_token → discard access_token

ACCESS TOKEN & REFRESH TOKEN — COMPLETE LIFECYCLE

Visual flow of how tokens are created, used, rotated, and expired

ONE-TIME SETUP (Manual)



AUTOMATED CYCLE (Repeating)



↳ New Refresh Token → Store securely

Token Type	Default TTL	Notes
Access Token	3600 sec (1 hour)	Customizable via scope: urn:opc:resource:expiry=<seconds>
Refresh Token	604,800 sec (7 days)	Maximum 7 days — CANNOT be extended. Rotate every 6 days.
Device Code	300 sec (5 minutes)	One-time use. Expires in 5 min if not verified in browser.
User Code	300 sec (5 minutes)	Must be entered in browser within 5 min of generation.

OAuth 2.0 — LIMITATIONS & CONSIDERATIONS

An honest assessment of OAuth challenges in Oracle EPM environments

Setup Complexity

- ⚠ Initial setup requires IDCS Administrator privileges — not self-service
- ⚠ First refresh token requires ONE-TIME manual browser interaction (Device Code Grant)
- ⚠ Separate OAuth clients required per application/connection in EDM
- ⚠ Understanding grant types, scopes & token flows requires training

Token Management Overhead

- ⚠ Refresh tokens expire after 7 days MAX — cannot be extended
- ⚠ If token expires (e.g., automation offline > 7 days), manual re-auth required
- ⚠ Token rotation must be automated — scheduled job every ≤6 days
- ⚠ Each refresh consumes old token — if new token not stored, access is lost
- ⚠ Encrypted .epw files must be managed per environment (dev/test/prod)

Functional Restrictions in EPM

- X EPM Automate: addUsers, removeUsers, assignRole, unassignRole STILL REQUIRE Basic Auth
- X OAuth login ignores username param — derived from token (can cause confusion)
- X Token must be generated by the same user who will execute REST API calls
- X Client Credentials grant (without user context) is NOT sufficient for EPM REST APIs

Common OAuth Errors to Handle

- **"invalid_grant" + "Token is expired"** → Refresh token > 7 days. Redo Device Code flow.
- **"invalid_grant" + "Token consumed"** → Old refresh token reused. Use latest only.
- **"invalid_request"** → Missing or invalid parameters in token request.
- **"invalid_scope"** → Scope string is incorrect or not authorized.
- **HTTP 401 Unauthorized** → Access token expired or invalid. Refresh and retry.

Despite these considerations, OAuth 2.0 remains the RECOMMENDED approach by Oracle. The operational overhead is manageable with proper automation.

EPM AUTOMATE — BASIC AUTH vs OAUTH LOGIN

Practical side-by-side comparison for automation scripts

Basic Auth Login

Command (plain text):

```
epmautomate login serviceAdmin P@ssword1  
https://epm-host.oraclecloud.com
```

Command (encrypted file):

```
epmautomate login serviceAdmin password.epw  
https://epm-host.oraclecloud.com
```

How .epw is created:

```
epmautomate encrypt PASSWORD KEY FILE
```

- ✗ Password stored in file (encrypted but static)
- ✗ Password changes break all scripts
- ✗ Not compatible with MFA
- ✗ Cannot use SSO credentials
- ✓ Simple to set up — minimal configuration
- ✓ Works for all EPM Automate configurations

OAuth 2.0 Login

Command:

```
epmautomate login ignoredUsername OAuth2.epw https://epm-host.ocs.oraclecloud.com
```

How .epw is created:

```
epmautomate encrypt REFRESH_TOKEN KEY FILE ClientID=CLIENT_ID
```

Username is DERIVED from token (param ignored)

- ✓ No password stored — only refresh token + client ID
- ✓ Token rotation provides ongoing security
- ✓ Compatible with federated identity (SAML + IDCS)
- ✓ Recommended by Oracle for all automation
- ✗ Cannot run: addUsers, removeUsers, assignRole
- ✗ Requires initial Device Code flow (one-time)
- ✗ Token must be refreshed every ≤6 days

Migration Checklist: Basic Auth → OAuth 2.0

- Register OAuth client in IDCS (one-time, Domain Admin)
- Obtain first refresh token via Device Code Grant
- Create new .epw file with ClientID parameter
- Update all scripts: replace password.epw with OAuth2.epw
- Set up scheduled token refresh job (every 6 days)
- Keep old Basic Auth .epw for user management commands
- Test in TEST environment before migrating PROD
- Update documentation and runbooks

DEMO 1: Getting YOUR FIRST ACCESS & REFRES TOKEN

One time setup

LIVE DEMO

Step 1- Post creating the OAuth2 in IDC

blue-moon-52210.postman.co/workspace/My-Workspace~c7e0478d-a871-4b85-82d0-10e9cae6f081/request/11291768-24c...

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My Workspace

Search

Invite Upgrade

POST Step 1- Post creating the OAuth2 in IDC

KSCOPE 26 - OAUTH Demo > Step 1- Post creating the OAuth2 in IDC

POST https://idcs-f928c0e1d2ea4601ad8e282f09b81c91.identity.oraclecloud.com/oauth2/v1/device?Content-type=application...

Docs Params Authorization Headers 9 Body Scripts Settings

none form-data x-www-form-urlencoded raw binary GraphQL

Key	Value	Description	Bulk Edit
response_type	device_code		
scope	urn:opc:idm:__myscopes__offline_access		
client_id	c70fb6356d8d4a6eb9f339fc412f6d9c		
redirect_uri	com.example.app/oauth2redirect&		
Key	Value	Description	

Body Cookies Headers 17 Test Results

200 OK • 718 ms • 928 B

Save Response

```
{
  "device_code": "16b9cbf2970c4c8bacbde9abcabb18a9",
  "user_code": "TBDLFGJF",
  "verification_uri": "https://idcs-f928c0e1d2ea4601ad8e282f09b81c91.identity.oraclecloud.com:443/ui/v1/device",
  "expires_in": 300
}
```

Connect Git Console Terminal

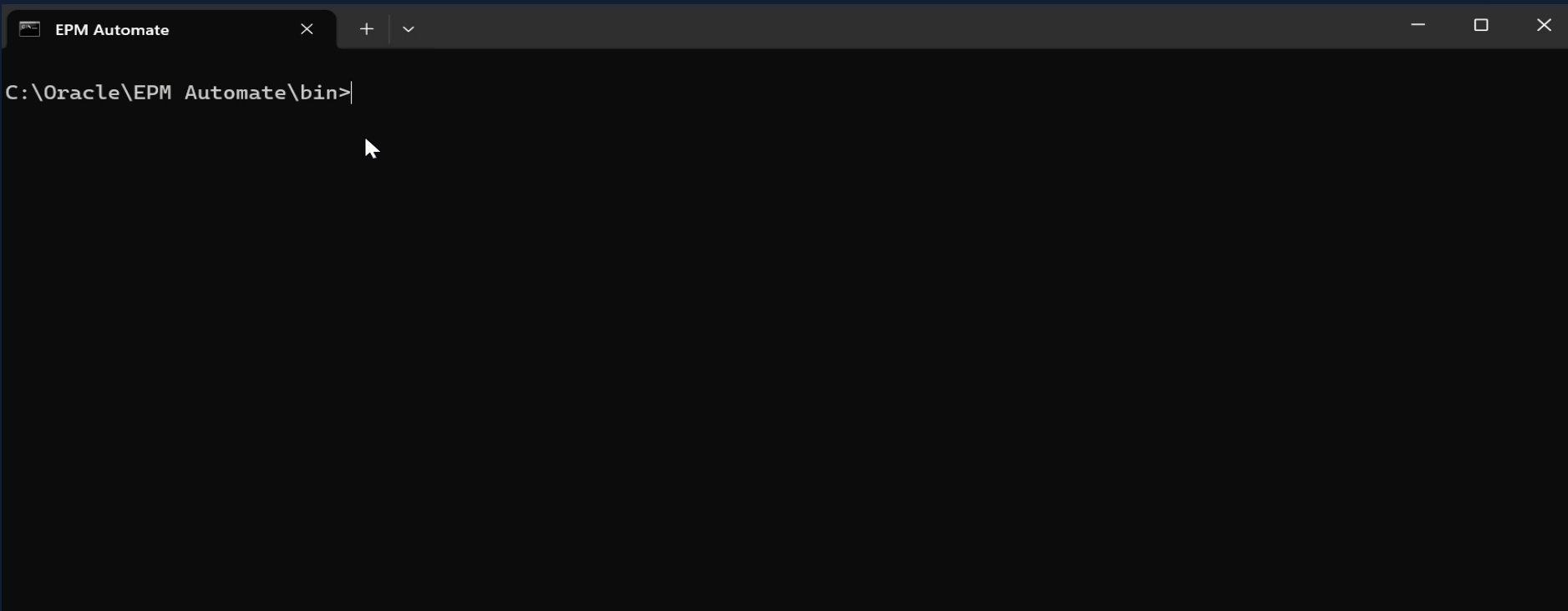
Cloud Agent Globals Vault Tools

4:18 PM 6/14/2026

DEMO 2: SECURE TOKEN-BASED ACCESS

LIVE DEMO

OAuth client credentials flow for EPM Automate



DEMO 2: SECURE TOKEN-BASED ACCESS

LIVE DEMO

OAuth client credentials flow for EPM Automate

EPM Automate



```
C:\Oracle\EPM Automate\bin>
```



DEMO 3: SECURE TOKEN-BASED ACCESS – REST API

[LIVE DEMO](#)

OAuth client credentials flow for EPM REST API authentication

KScope 26 - EPM13 > Get Job Definitions - OAUTH 2.0

GET https://epm13-test-a641377.epm.us-phoenix-1.ocs.oraclecloud.com/HyperionPlanning/rest/v3/applications/EPBCS/jobde... Send

Docs Params Authorization Headers 11 Body Scripts Settings Cookies

Headers 8 hidden

Key	Value	Description	Bulk Edit	Presets	...
☒ Content-type	application/json				
☒ Authorization	Bearer eyJ4NXQjUzI1Ni6lnF0SGpDUGo5cndmT...				
☐					

Body Cookies 2 Headers 18 Test Results 200 OK • 689 ms • 22.95 KB • Save Response

{ JSON Preview Visualize

```
1 {
2   "items": [
3     {
4       "jobName": "RefreshCube",
5       "jobType": "Cube Refresh"
6     },
7   ]
8 }
```

No Passwords Stored

Client ID & secret stored as environment variables or vault secrets — never in code.

Automatic Expiry

Tokens expire after configured TTL (e.g., 1 hour). No manual rotation needed.

Use in API Headers

Authorization: Bearer <token> — works across all EPM REST endpoints.

CROSS-PRODUCT INTEGRATION – Use Case

Seamless workflows across Planning, Financial Consolidation, and FreeForm using a single OAuth token

Oracle Planning

/epm/rest/v1/applications

Key Operations:

- Data Load / Rules
- Scenario Management
- Dimension Updates
- Approvals Workflow

Fin. Consolidation

/epm/rest/fccs/v1/

Key Operations:

- Consolidation Run
- Data Status Check
- Journal Management
- Report Trigger

FreeForm Cloud

/epm/rest/v1/freeform/

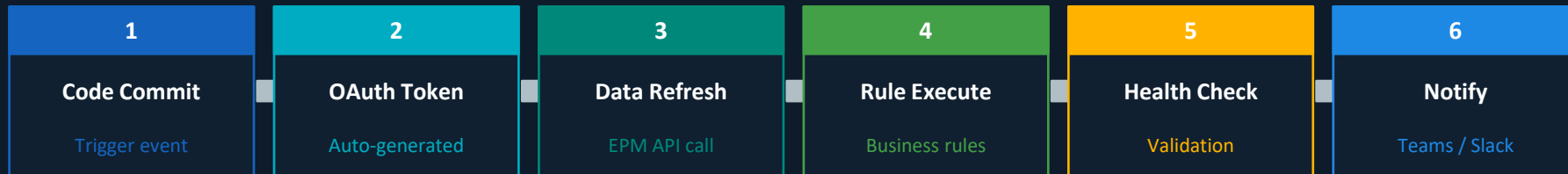
Key Operations:

- Custom Model Jobs
- Business Rules
- Data Forms API
- Metadata Sync

Single OAuth token. Three products. One secure, unified pipeline.

AUTOMATED CI/CD PIPELINES – Use Case

OAuth-powered automation for data refresh, scenario management & health monitoring



Pipeline Use Cases:

Nightly Data Refresh

Trigger EPM data loads automatically at midnight using OAuth client credentials in a scheduled pipeline job.

Scenario Automation

Create, clone, and manage Planning scenarios via REST APIs — all authenticated with scoped OAuth tokens.

Health Monitoring

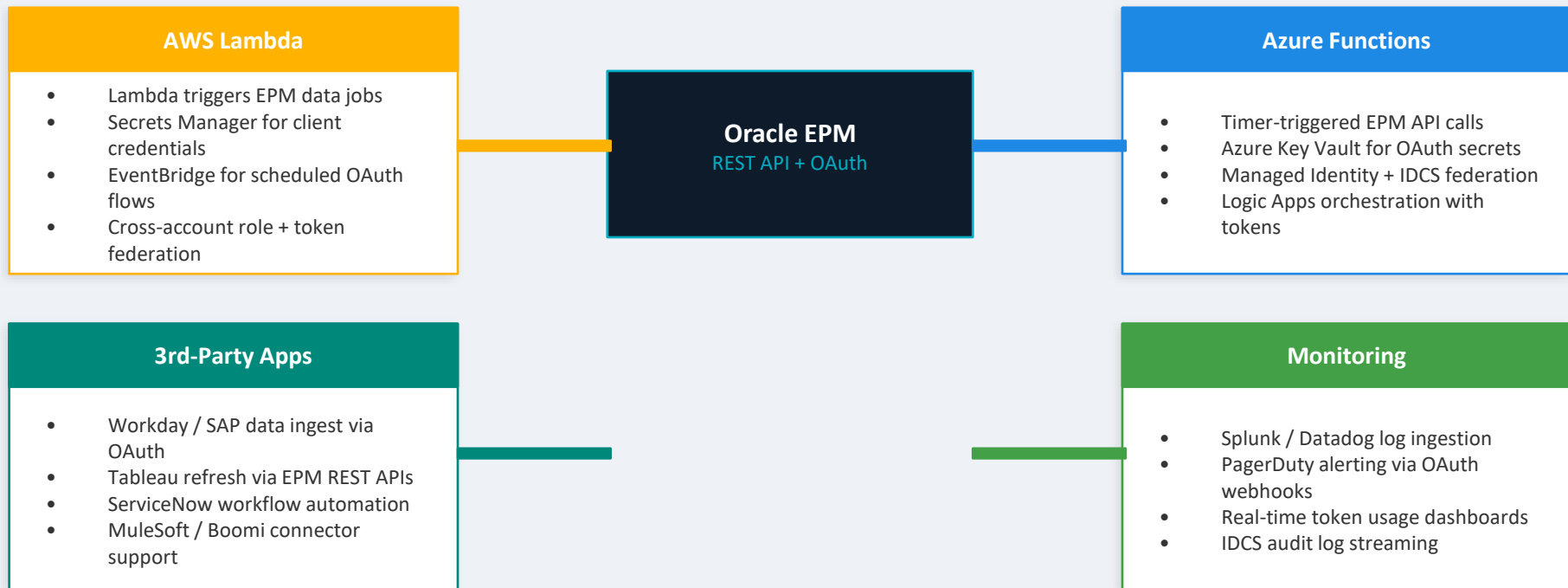
Ping EPM REST health endpoints every 5 min. Alert on failures via OAuth-authenticated webhook calls.

Release Deployment

Deploy metadata changes, dimension updates and business rules across environments using CI/CD + OAuth.

MULTI-CLOUD CONNECTIVITY – Use case

Integrating Oracle EPM securely with AWS Lambda, Azure Functions, and third-party services



IDCS & OAUTH GOVERNANCE

Oracle Identity Cloud Service — Managing roles, scopes & token lifecycles for enterprise compliance

Role & Scope Management

- Define granular OAuth scopes per EPM product (Planning.Admin, FCCS.Read, etc.)
- Map IDCS application roles to EPM business roles
- Enforce least-privilege access per integration client
- Segregation of duties via scope assignments

Token Lifecycle Control

- Configure token TTL (15min to 1hr for automation)
- Refresh token rotation policies
- Immediate revocation for terminated accounts
- Token introspection endpoint for validation

Audit & Compliance

- Full audit log of token issuance & usage
- SIEM integration (Splunk, QRadar)
- SOC 2 / ISO 27001 alignment
- Scheduled access reviews via IDCS governance

Multi-Tenant Security

- Isolated OAuth clients per environment (dev/qa/prod)
- Cross-domain federated identity (SAML + OAuth)
- IP allowlisting for client credentials
- mTLS for high-security API clients

Implementing OAuth in Real Pipelines



OAuth Client Registration

Registering an OAuth client in Oracle Identity Cloud Service sets the base for secure token issuance.

Secure Token Storage

Store refresh tokens and client IDs securely using encryption or secret management tools like Azure Key Vault.

Automated Token Retrieval

Scripts automate retrieval of access tokens using refresh tokens to authenticate API requests without passwords.

Pipeline Deployment & Best Practices

Deploy pipelines across environments with dedicated accounts, segregation, and continuous monitoring for security.

KEY TAKEAWAYS

01

OAuth is the Modern Standard

OAuth 2.0 replaces Basic Auth with token-based access — eliminating hard-coded credentials, enabling scope control, and meeting enterprise compliance requirements.

02

Step-by-Step Implementation

Implement OAuth tokens in automation scripts using IDCS client credentials flow. Works across Oracle Planning, FCCS, FreeForm, and all EPM REST endpoints.

03

IDCS Governance is Key

Use Oracle IDCS to manage scopes, roles, and token lifecycles. Monitor usage, revoke instantly, and maintain full audit trails for SOC 2 compliance.

04

Multi-Cloud Ready Architecture

OAuth enables seamless EPM integration with AWS Lambda, Azure Functions, and third-party tools — building a truly hybrid, cloud-native EPM ecosystem.

05

Upgrade Existing Integrations

Migrate existing Basic Auth scripts to OAuth safely using the client credentials grant type. Minimal code changes — maximum security and scalability gains.

Start Building Secure EPM Pipelines Today

Modernize • Automate • Secure

- 1 Register an OAuth client in Oracle IDCS today
- 2 Replace your first Basic Auth script with OAuth tokens
- 3 Implement token scoping for each EPM product
- 4 Build your first multi-cloud EPM integration pipeline

Track

**EPM Master Data
& Integrations**

Format

Demo-Driven
Technical Session

Level

Intermediate
to Advanced



Thank you for your time today.



The presentation and POSTMAN extracts have been made available at: <https://github.com/pdayalan/Kscope26>